

Tirth Patel

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EDUCATION

University of Regina

Expected Graduation Aug 2028

BSc in Computer Science, Co-op Program

- **Coursework:** Data Structures & Algorithms, Database Systems, Web Programming, Software Engineering, Operating Systems

PROJECT EXPERIENCE

AWS CloudTrail Security Monitoring: Detecting Suspicious IAM Activity

[Github Repo](#)

Tech Stack / Skills: AWS CloudTrail, Amazon EventBridge, Amazon Simple Notification Service (SNS), Amazon S3, AWS Identity and Access Management (IAM), MITRE ATT&CK, JSON, Incident Response

- Built a **real-time cloud threat detection** pipeline to test whether leaked AWS credentials used to plant a backdoor admin and disable logging would be noticed
- Configured a multi-region **AWS CloudTrail** trail archiving to **Amazon S3** and wrote 3 **Amazon EventBridge** detection rules mapped to **MITRE ATT&CK** (T1136.003, T1098, T1562.008, T1078.004), routing matches to **Amazon SNS** email alerts
- Simulated a stolen-credential attack on my own account; **4 of 5 attack steps triggered alerts within seconds**
- Investigated the missed root login in CloudTrail Event History, traced it to a **cross-region blind spot**, and documented the timeline and MFA gaps in an **incident report**

Cerberus: Threat Monitoring Dashboard

[Github Repo](#)

Tech Stack / Skills: Python, FastAPI, SQLAlchemy, JWT, dnspython, React, Vite, Tailwind CSS, SQLite, Docker, Docker Compose, REST API

- Developed a self-hosted **threat intelligence dashboard** that checks IP and domain reputation in one place, replacing one-at-a-time blacklist lookups
- Engineered a **FastAPI** backend that scans **60+ DNSBL providers**, pulls **AbuseIPDB** scores and runs WHOIS, DNS, SSL and **SPF/DKIM/DMARC** checks
- Hardened the app with **JWT authentication**, per-client **rate limiting**, **SSRF protection** and secure-by-default configuration
- Shipped a **React** frontend with scheduled monitoring, email and webhook alerts and history charts, containerized with **Docker Compose** over 5 releases

Flappy Bird: Left to Right and Back

[Github Repo](#)

Tech Stack / Skills: Python, pygame, Git, GitHub

- Built a twist on the classic game in **Python** and **pygame** where the bird reverses direction every 5 points and flies back through the pipes it just passed
- Implemented a **60 FPS game loop** with gravity, **collision detection** and random pipes, reversing the scrolling world by flipping one direction variable
- Drew every sprite in code with no image assets, adding pause, restart and a high score saved to file

PROFESSIONAL EXPERIENCE

Technology Intern, Tech Department

Mar 2024 – Jul 2024

Kanan International Private Limited (Kanan.co) | Full-time Internship

Vadodara, Gujarat, India

- Delivered **end-user technical support** to faculty during live online classes, resolving portal issues in real time
- Provisioned user access to the portal with **CRM** and **user authentication tools**, supporting **access management**
- Organized the portal video database and supported the **K-APPLY** and **KYS** platforms using **Microsoft Office**

OTHER EXPERIENCE

Crew Trainer

Oct 2024 – Mar 2026

McDonald's Canada | Part-time, Permanent

- Trained and onboarded **25+ new crew members** on standard operating procedures in a high-volume environment

Produce Associate | Cross-Trained: Deli, Bakery, Meat

Jun 2025 – Present

Walmart | Part-time, Permanent

- Supported daily operations across **Produce, Deli, Bakery and Meat**, handling product rotation, merchandising, **food-safety compliance** and waste control

CERTIFICATIONS

- **Cybersecurity Job Simulation (Phishing) – Mastercard, Forge**
- **Ethical Hacking – MSME-Technology Development Centre, Government of India**

Sep 2026

Jun 2023